

Arya Rane

Curriculum Vitae

Bachelor of Chemical Engineering
Institute of Chemical Technology, Mumbai
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Research Experience

Indian Institute of Technology, Bombay

- May, 2026 – **Overlapping Graph Decomposition for Fault-Tolerant Distributed MPC.**
present
 - Formulated an overlapping network decomposition scheme for integrated process plants via Genetic Algorithm on weighted equation graphs.
 - Implemented an iterative Distributed MPC coordinator in Python using `do_mpc` and IPOPT, resolving coupled nonlinear subproblems via trajectory broadcasting and input-freezing protocols.
 - Validated dynamic fault tolerance on a multiple-recycle benzene alkylation benchmark, achieving zero tracking degradation (identical ISE) with a 30.8% solve-time reduction under sub-controller failure.

Advisor : **Prof. Sujit Jogwar**, Associate Professor, Department of Chemical Engineering, IIT Bombay.

Institute of Chemical Technology, Mumbai

- Sept, 2025 – **Adaptive Neural Surrogates & Inverse Optimization for Dynamical Systems.**
present
 - Built neural network surrogate models using error-driven adaptive spatial sampling, applying quantile-based error flagging and PCA-aligned subgrids to resolve steep response gradients with minimal training points.
 - Implemented Physics-Informed Neural Networks (PINNs) in PyTorch to solve forward dynamics and inverse parameter discovery (NSE flow fields) via automatic differentiation of differential residuals.
 - Formulated a gradient-driven inverse optimization framework utilizing Moore-Penrose pseudoinverse to optimize inputs directly on learned neural landscapes.

Advisor : **Prof. Ajit Kumar**, Professor & Head, Department of Mathematics, ICT Mumbai
Dr. Shailesh Kumar, Chief Data Scientist, AI/ML Centre of Excellence; Dean, Jio Institute, Reliance Jio

Dec, 2024 – **Multiphase CFD.**
July, 2025
 - Studied fluid mechanics, turbulence modeling and CFD for incompressible single- and multiphase systems, and performed numerical simulations using OpenFOAM.
 - Simulated transient bubble hydrodynamics and interfacial dynamics for immiscible systems (benzene/toluene rising in water) using the Volume of Fluid (interFoam) method in OpenFOAM.

Advisor : **Prof. Ashwin W. Patwardhan**, Senior Professor, Department of Chemical Engineering, ICT Mumbai

Research Interests

Applied Mathematics, Process Systems Engineering, Fluid Mechanics, Optimization.

Education

- 2023 – 2027 **Institute of Chemical Technology (ICT)**, Mumbai, India.
Bachelor of Chemical Engineering
Minor: Artificial Intelligence & Machine Learning
Undergraduate Research Thesis (Advisor: Prof. G. D. Yadav): — Valorization of CO₂ to lower olefins via One-Pot Bifunctional Catalysis
CGPA: 8.23/10

Academic Achievements

- All India Rank **76** Graduate Aptitude Test in Engineering (GATE) 2026.
76 Chemical Engineering CH — among 14070 ChemE students across India
370 Engineering Sciences XE: Engineering Mathematics, Fluid Mechanics, Thermodynamics

2025 **iGEM Competition 2025 — ICT Mumbai won Gold Medal.**

Co-developed DNA-damage repair (Belov model) model translation & programming.

Skills

Programming	Python (PyTorch, NumPy, SciPy, CasADi), C++
Scientific Computing	Numerical methods, optimization, Neural Networks, CFD, HPC
Tools	OpenFOAM, Linux, Bash, Git, $\LaTeX$, DWSIM, Simulink

Selected Academic Projects

Thermodynamics	Built a Python program for TAC minimization vs. operating pressure for a binary distillation column , using NRTL-regressed VLE data.
Data Analytics & ML	Performed exploratory data analysis and regression-based modelling of geospatial temperature data for climate trend analysis. Conducted statistical, and spatial analysis of geochemical data using unsupervised learning techniques.
Optimization	Implementation of direct search methods using SageMath . Studied and implemented Sequential Quadratic Programming (SQP) and its variants. Implementation and benchmarking of L-BFGS method in Python.

Extra-Curricular Activities

- Jan 2026 **Gas-Inducing Impeller Design Competition – Tinkerer’s Lab, ICT Mumbai.**
- Secured 1st place among 7 teams; designed and fabricated a 92 mm literature-derived gas-inducing impeller (CAD, 3D printing, laser cutting).
 - Achieved gas induction onset at $N_{cg} = 368$ RPM and solid suspension ($d_p \approx 2\text{--}3$ mm) at 500 RPM, delivering a mass-transfer-to-power efficiency of $1.02 \times 10^{-5} \text{ mol L}^{-1}\text{s}^{-1}\text{W}^{-1}$ ($1.8\times$ over runner-up) and an average $\text{CO}_2\text{--NaOH}$ mass transfer rate of $1.26 \times 10^{-3} \text{ mol L}^{-1}\text{s}^{-1}$ at 123 W.
- Jan 2026 **Codathon – Vortex 13.0 Tech Fest, ICT Mumbai.**
- Designed the problem statement, coordinated evaluation and logistics for the coding competition.